



A modified social force model for sudden attack evacuation based on Yerkes–Dodson law and the tendency toward low risk areas

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ABSTRACT

Various evacuation models are the internal support for the movement of pedestrians. Emergencies that occur in public places have the characteristics of suddenness, massive and unpredictability. These characteristics impact pedestrians' psychological states, ultimately shaping their behaviors. However, the existing models do not fully consider this aspect. Therefore, for sudden attack events, based on the social force model and Yerkes–Dodson law, this paper proposes a modified model which considers the positive and negative effects of psychological state on pedestrians' behaviors in three dimensions: the realization degree of desired speed, the proximity of attacker to pedestrian, and crowdedness. It also considers the diminishment of repulsion and the attractiveness of low risk areas. A behavioral model based on the exit-reorientation mechanism is proposed to describe the behavior of low risk exit selection according to the relative location of the attacker. The improved model is validated quantitatively and qualitatively, respectively. Simulations based on this model can effectively reproduce the behaviors of crowds in videos. The data of fundamental diagrams from our model fits well with that from videos. We find that the evacuation time is longer when the observation distance within a specific range in the simple scenario. The risk tolerance level of 8 m and 6 m are the critical points affecting the rate of change of survival rate and casualties, respectively. Simulation results indicate that the effect of rational people ratio on evacuation time will vary with risk tolerance level. This study offers insights into the dynamics of crowd evolution during an evacuation following a sudden attack, providing a foundation for exit allocation and emergency crowd guidance.

1. Introduction

With the rapid development of the economy, the acceleration of urbanization, and the rapid increase of population density, more and more people gather in public places for work, entertainment, or participating in events. A sudden attack could cause enormous casualties without awareness of crowd evolution and appropriate crowd management. On January 1, 2010, a suicide car bomb attack occurred in a volleyball court in Pakistan, killing at least 95 people and injuring more than 100 people; On August 18, 2017, a knife attack occurred in Turku, southwest Finland. On May 24, 2022, an 18-year-old high school student carried out a shooting rampage at Robb Elementary School in the United States, resulting in 21 fatalities and at least 17 injuries. On November 25, 2022,

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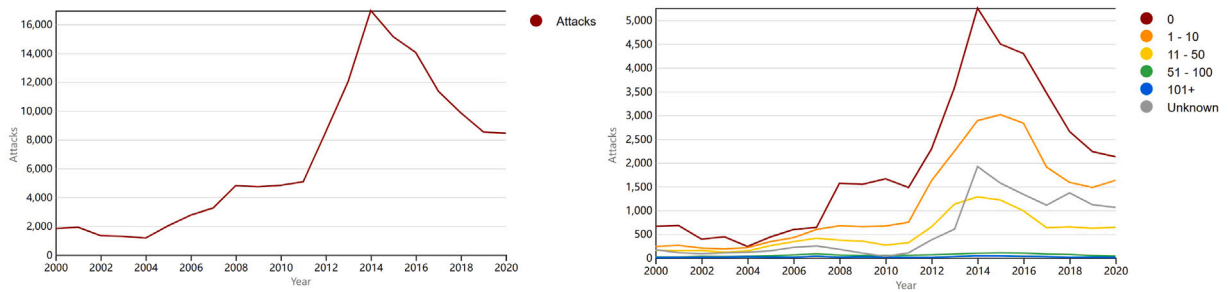


Fig. 1. The number of attacks and casualties worldwide in recent years [4].

two schools in Brazil experienced a shooting incident that resulted in 4 fatalities and 12 injuries. Terrorism is becoming a significant threat to the world's peace with an international trend. From GTD global terrorism database, the number of attacks and casualties worldwide in recent years is shown in Fig. 1.

Therefore, sudden attacks in public places are a significant threat to people's lives and property. Investigating the interactions between pedestrians, attackers, and environment, as well as how these interactions lead to various crowd phenomena, holds significant importance. Many researchers have already undertaken extensive work in this field. The purpose of conducting this research is to apply its findings toward the establishment of more effective emergency management strategies for sudden attacks.

To explore the crowd evacuation features, various kinds of models are built. In general, evacuation dynamics models are divided into three or two categories: macroscopic, mesoscopic, and microscopic models or continuous and discrete models depending on the properties of the models. Handerson [1] is the first to propose the macroscopic model, which considers pedestrian flow as an approximation to the gas flow or fluid. Macroscopic models are rooted in hydrodynamic models, which are difficult to describe the complex pedestrian behavior observed in real life. The lattice gas model [2], as a representative of the mesoscopic model, integrates the properties of macroscopic and microscopic models, dividing the scenario into small grids where pedestrians have mass but no volume and can only move along the grid lines. Although its research scale is at the level of pedestrians, it only considers the interactions between pedestrians a little, which seems rougher than the microscopic model. Cellular automaton is a microscopic discrete model in which time, space, and state are discrete with the advantage of high calculative efficiency. It uses a series of model-constructed rules to determine the evacuation routes of pedestrians and states of cells. The queueing network model is another discrete microscopic model. It does not focus on the structure of the building and cannot describe the particular behavior of pedestrians in detail, but its computation speed is fast. The social force model is proposed by Helbing et al. [3], which is a classic microscopic continuous model. Three types of forces drive pedestrians in this model. These forces describe interactions between people and obstacles physically and psychologically. In order to investigate crowd evacuation under sudden attack better, researches based on these models with considering characteristics of events and pedestrians have been continuously conducted.

Firstly, research on behavior and strategy modeling has received considerable attention. Extracting abnormal behaviors based on video is one of the methods of investigating attack evacuation. Liu et al. [5] put forward a calculation method of pedestrian flow Reynolds number in the video. It could be applied to pattern analysis of crowd abnormal behavior analysis, which can be used to improve the efficiency of public security management. Kang et al. [6] propose a crowd abnormal detection algorithm based on a modified social force model (MSF) to recognize the abnormal behavior of panics and flees caused by emergencies in public. This study could provide decision support for emergency response in sudden attacks. Wang et al. [7] use particle image velocimetry based on cross-correlation algorithm to statistically analyze crowd velocity in video surveillance under a terror attack. Several phenomena are found, including the crowd oscillating phenomenon and the self-organizing grouping phenomenon, which will benefit the understanding of crowd dynamics in close-range attacks. In terms of the impact of behavior and strategy on evacuation outcomes, Young et al. [8] indicate that faster evacuation and higher traffic volumes with greater congestion at exits are observed during high motivation scenarios. Zhang et al. [9] develop a multi-agent simulation system and apply it in a simplified violent attack scenario. They analyzed attackers' target search strategies and formulated civilians' exit selection behavior based on a discrete selection model. They also proposed a civilian behavior model based on perception-decision-behavior framework and terrorist behavioral model with the finite state machine, which can be used to study the relationship between civilian characteristics and event consequences [10]. Deng et al. [11] explore the improving influence of information sharing on indoor emergency evacuation performance. Cho et al. [12] investigate the effect of sensing density coverage on the level of casualties. Liu [13] innovatively introduced an algorithm simulating the fight process between security guards and attackers, providing a detailed analysis of the effect of security guards' defensive strategies on casualties. Regarding attackers' behaviors, Bernardini G et al. [14] innovatively proposed a behavioral database for attacker acts according to consolidated methods. They highlight differences between behavioral outputs of qualitative and quantitative variables considering attacker acts and other evacuation types. It proves that their data could be used as input in the context of attacker acts. Moreover, agent-based models of terrorist behavior presented by Elliott et al. [15] are recommended as policy analytic tools.

Secondly, an abundance of improved crowd dynamics modeling research has emerged. These models are designed to simulate crowd evacuation in high-risk scenarios, providing a foundation for emergency management. Yu et al. [16] build a three-dimensional cellular automata model containing the speed of moving up and down the stairs, position attraction, and probability of collisions. It

helps to make a reasonable evacuation plan for the crowd under sudden attack. Li et al. [17] establish a driving-forces model based on the social force model and apply it in the scenario, including moving threat agents. This model considers the instinctive and strategic driving forces caused by threatening subjects. It can be used to study the evacuation of an attack with close-range weapons in public places. Liu [18] investigates the evacuation under terrorist attacks in public places with an improved social force model and finds the relationship between the starting positions of terrorists and the number of casualties. Moreover, some improvements in models focus on panic and its derivative effects. The pedestrian evacuation model proposed by Zhang et al. [19] considers the promotion of panic to speed. Zheng et al. [20] propose the concept of emergency diffusion field to describe the diffusion effect of emergency on pedestrian evacuation process based on the floor field model, and the panic coefficient proposed can describe the influence of panic on evacuation process. Chen et al. [21] propose an extended floor field cellular automata model considering pedestrians' panic. Wang et al.'s research shows that people tend to ignore the effective exit in panic. If a certain number of leaders guide the evacuation, the evacuation efficiency will be greatly improved [22]. It also provides a basis for the emergency management of security personnel in the attack.

Currently, simulation based on models has become increasingly popular for pedestrian dynamics investigation in various scenarios. Despite several well-developed evacuation models, the social force model, as one of the continuous microscopic models, has the advantage of describing the movement and interactions of people through the physical world. Since people are in different physical and psychological situations, an appropriate model should consider the condition heterogeneity under the sudden attack evacuation. For example, people close to attackers primarily aim to escape from the area where attackers can hurt them. At the same time, those who are relatively safer choose to flee directly to the exits. That is to say, the simulation should depict the picture of all kinds of people's states, so a macro model that considers everyone as the same individual is not recommended. Cellular automata model as a micro model has the feature of high computational efficiency, but it also has limitations of detail description of attack evacuation simulation. Therefore, this paper chooses the social force model as the basis of a modified model for the evacuation simulation of pedestrians under a sudden attack.

To solve the issues mentioned above, we combine psychological factors with the model. The source which motivates the generation of self-driving force in most current modified social force models is the gap between the desired speed and actual speed, which considers that the more significant the gap, the greater the self-driving force [3]. The Yerkes-Dodson law, proposed by psychologists, suggests that excessive psychological stress can reduce people's performance. Therefore, we intend to explore the evolution of crowd evacuation under sudden attack by integrating Yerkes-Dodson law in the social force model with the nonlinear effect of the gap between desire and reality on the desired speed in this paper.

2. Model

2.1. Basic social force model

The social force model [3] uses Newton's second law to describe the interaction between people and the external environment. It is assumed that among N pedestrians, the mass of each pedestrian i is m_i . Self-driving force describes the phenomenon that pedestrians want to move in a specific direction e_i^0 at a desired speed v_i^0 , and tend to adjust their actual speed v_i within the characteristic time τ_i . At the same time, pedestrians try to keep a distance from other pedestrians j and walls W , which can be expressed by repulsive forces f_{ij} and f_{iW} , respectively. The formula is as shown in Eq. (1).

$$m_i \frac{dv_i}{dt} = m_i \frac{v_i^0(t) e_i^0(t) - v_i(t)}{\tau_i} + \sum_{j(\neq i)} f_{ij} + \sum_W f_{iW} \quad (1)$$

2.2. Improved social force model

In order to make sudden attack evacuation simulation more realistic, we propose an improved model considering the pedestrians' heterogeneity of physical and psychological conditions by combining the social force model with the Yerkes-Dodson law and the tendency toward low risk areas.

2.2.1. Positive and negative effects of pressure on pedestrians

The improved model considers pressure's positive and negative effects on pedestrians' desired speed in three dimensions. Many researchers have done substantial experiments to observe and analyze people's responses in unexpected situations and have found some relationship between pressure and performance. Yerkes-Dodson law is an empirical relationship between pressure and performance, originally developed by psychologists Robert M. Yerkes and John Dillingham Dodson in 1908 [23]. As depicted in Fig. 2, performance improves with physiological or mental arousal, but only up to a certain threshold. When arousal levels exceed this threshold, performance declines. Therefore, an appropriate level of mental arousal can promote the pedestrians' ability to cope with the emergency. Mental arousal can reflect the level of panic to a certain extent.

In Ref [3], mental arousal is determined only by the realization degree of desired speed, and it always promotes desired speed increasing. Many factors can make pedestrians panic during an evacuation, and they will not always promote desired speed increases. On the contrary, people will too panic to speed up according to the Yerkes-Dodson law. Therefore, the improved model considers two other factors except for the realization degree of desired speed: the proximity of the attacker to pedestrians and crowdedness, respectively, drawing inspiration from the format of original self-driving force [3]. This paper proposes an improved

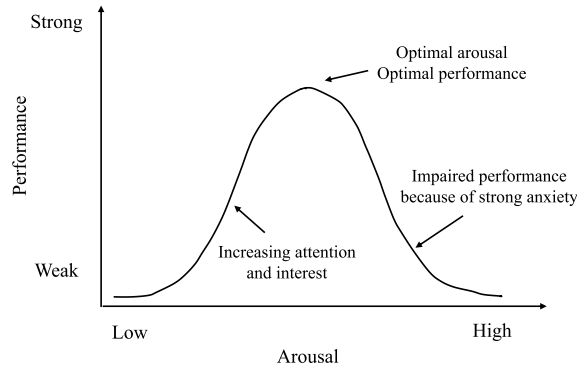


Fig. 2. Yerkes-Dodson law curve [24].

time-psychology-dependent parameter q as Eq. (2). Herein, a , b , and c are positive constants between 0 and 1, and $a + b + c = 1$. Desired speed updates based on Eq. (3). $g_1(x)$, $g_2(x)$, $g_3(x)$, n_{1i} , n_{2i} and n_{3i} are explained in the following.

$$q = ag_1(n_{1i}) + bg_2(n_{2i}) + cg_3(n_{3i}) \quad (2)$$

$$v_i^0(t) = v_i^{\min} + q(v_i^{\max} - v_i^{\min}) \quad (3)$$

(1) Realization degree of desired speed

The panic parameter n_{1i} of pedestrian i caused by the realization degree of desired speed can be described by the following Eq. (4).

$$n_{1i} = \frac{|\mathbf{v}_i^0(t) - \mathbf{v}_i(t)|}{v_i^0(t)} \quad (4)$$

Herein, $v_i^0(t)$ is the magnitude of the desired speed of pedestrian i at time t , and $\mathbf{v}_i(t)$ denotes the actual speed at time t . $\mathbf{v}_i^0(t)$ represents the vector of desired speed. How the panic parameter n_{1i} contributes to the speed is determined by the Eq. (5) where x represents the realization degree of desired speed, i.e., n_{1i} . An appropriate degree of difference between the desired and actual speeds can encourage pedestrians to speed up. However, too much difference could make a negative influence.

$$g_1(x) = \begin{cases} x, & 0 \leq x \leq 0.65 \\ -x, & 0.65 < x \leq 1 \\ -1, & x > 1 \end{cases} \quad (5)$$

(2) Proximity of attacker to pedestrian

The proximity of the attacker to the pedestrian as a significant threat in a sudden attack evacuation dramatically affects pedestrians' panic. The panic parameter n_{2i} of pedestrian i caused by the proximity degree of the attacker to the pedestrian can be described by the following Eq. (6).

$$n_{2i} = \frac{d_{act} - d_{ia}}{d_{act}} \quad (6)$$

Here, d_{act} is the minimum distance for risk tolerance. Before the distance between the attacker and pedestrian d_{ia} is closer than d_{act} , pedestrians will flee to the exits regardless of where the attacker is. Similarly, $g_2(x)$ describes the positive and negative effects on pedestrians caused by the proximity of the attacker to the pedestrian as shown in Eq. (7).

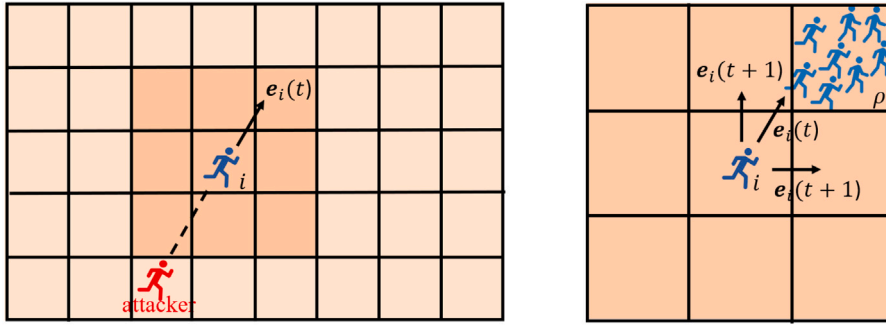
$$g_2(x) = \begin{cases} 0, & x < 0 \\ x, & 0 \leq x \leq 0.65 \\ -x, & x > 0.65 \end{cases} \quad (7)$$

(3) Crowdedness

When a pedestrian finds a crowd of people in front of him at a critical moment of evacuation, will he accelerate to the front crowd out of panic? We propose a parameter n_{3i} to describe the panic caused by crowdedness.

$$n_{3i} = \frac{\rho - \rho_{immovable}}{\rho_{immovable}} \quad (8)$$

Here, ρ is the crowd's density in the direction of the desired speed of pedestrian i . Furthermore, $\rho_{immovable}$ denotes the critical density that people are too crowded to move. If the crowd's density is within a specific range, the pedestrian will accelerate and bypass the crowd, while if it is too crowded ahead, the pedestrian will slow down, wait or change direction. We divided the



$\rho_{immovable}$ is a constant.

- If $\rho \leq \rho_{immovable}$, the crowd can accommodate more people
- If $\rho > \rho_{immovable}$, The crowd will have difficulty moving, pedestrian i may choose to change his/her desired direction.

Fig. 3. The impact of crowd density on the desired direction.

entire scene into rectangular regions for calculating the crowd density ρ in each region. Fig. 3 illustrates the potential impact of crowdedness on the desired direction. $g_3(x)$ describes this change in behaviors.

$$g_3(x) = \begin{cases} 0, & x < 0 \\ x, & 0 \leq x \leq 0.65 \\ -x, & 0.65 < x < 1 \\ -1, & x \geq 1 \end{cases} \quad (9)$$

2.2.2. Panic leads to a diminishment of repulsion

With the influence of panic, the repulsive forces between pedestrians and obstacles or other pedestrians are likely to diminish [25]. Based on the above three dimensions of improvements, we propose a panic parameter w to describe the panic emotion of pedestrians. This parameter affects the repulsive forces, as shown in Eq. (11). $v_i^0(t)$ and $v_i(t)$ hold the same meaning as defined in Eq. (4). τ_i represents the characteristic time, f_{ij} denotes the inter-individual repulsion, and f_{iW} signifies the repulsion exerted by obstacles on individuals. F_{is} represents the attractiveness of low-risk exits. The explanation on how to determine low-risk exits and calculate the attractiveness will be provided in subsequent sections.

$$w = a |g_1(n_{1i})| + b |g_2(n_{2i})| + c |g_3(n_{3i})| \quad (10)$$

$$m_i \frac{d\mathbf{V}_i}{dt} = m_i \frac{v_i^0(t) \mathbf{e}_i^0(t) - \mathbf{v}_i(t)}{\tau_i} + \sum_{j(\neq i)} (1-w) \mathbf{f}_{ij} + \sum_W (1-w) \mathbf{f}_{iW} + \mathbf{F}_{is} \quad (11)$$

2.2.3. Exit-reorientation behavior

Pedestrians tend to change their initial route choice decision in crowd evacuation [26]. During the sudden attack evacuation in a multi-exit room, pedestrians may redetermine the target exit to approach the low risk exit based on the relative position between the attacker, exits, and themselves. If the attacker is on the route to the target exit, that exit is considered a high-risk exit. Therefore, the exit-reorientation mechanism prevents pedestrians from fleeing to the closest exit, meanwhile closer to the attacker, which is shown in Fig. 4. Here, $\mathbf{n}_{ie_1}$ is the unit vector from the pedestrian to the target exit, $\mathbf{n}_{ia}$ is the unit vector from the pedestrian to the attacker. $d_{observe}$ is the observation distance. It also affects the reorientation process. For example, in Fig. 5(a), the target exit of pedestrian i is the nearest exit e_1 , and the attacker is in the potential route of pedestrian i judging according to the exit-reorientation mechanism. Since exit e_2 satisfies $\mathbf{n}_{ie_2} \cdot \mathbf{n}_{ia} < \mathbf{n}_{ie_1} \cdot \mathbf{n}_{ia}$ and no exit satisfies $\mathbf{n}_{ie_1} \cdot \mathbf{n}_{ia} < 0$, the target exit of pedestrian i is updated to e_2 . Similarly, pedestrian i in Fig. 5(b) keeps the current target exit e_1 , and pedestrian i in Fig. 5(c) updates the target exit to e_3 which satisfies $\mathbf{n}_{ie_3} \cdot \mathbf{n}_{ia} < 0$. We set the parameter $d_{observe}$ because what pedestrians can observe is related to the radius of vision. Peijie Ma et al.[27] studied the relationship between evacuation time and the radius of vision. Li et al.[28] also studied the influence of visual radius and psychological tension on evacuation by introducing a new visibility function.

2.2.4. Attractiveness of low risk area

When pedestrians are in the same space as the attackers, the space near obstacles that provide shelter is safer and less risky. However, when the room has few obstacles, escaping to the space outside the room will be a safer choice. The space outside the room can be seen as a low risk area, with walls as the obstacles. Therefore, the exits between attacker space and low risk area

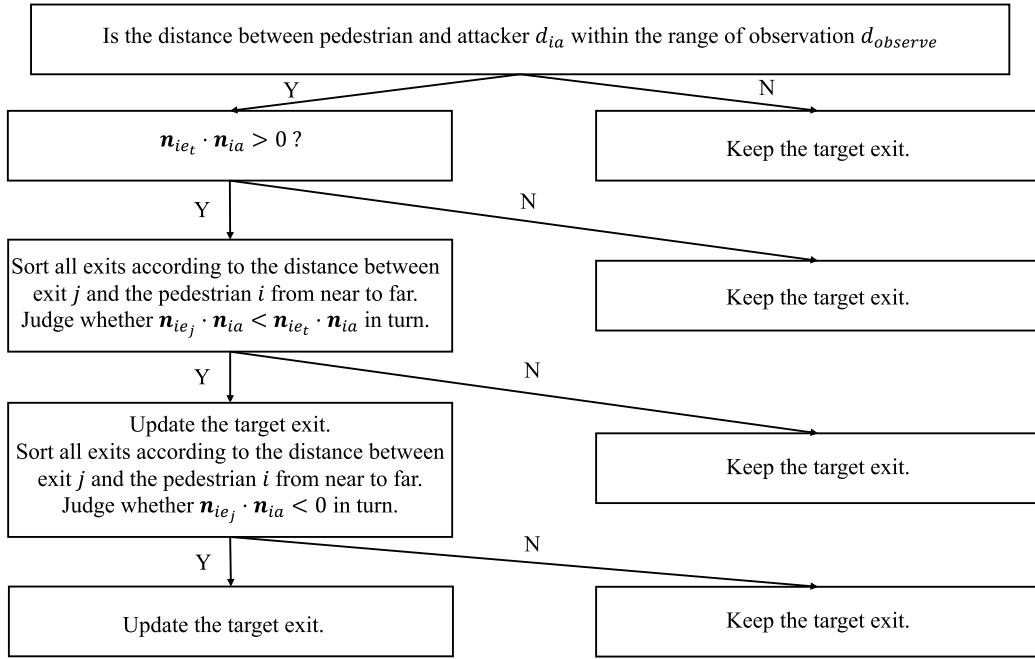


Fig. 4. The exit-reorientation mechanism.

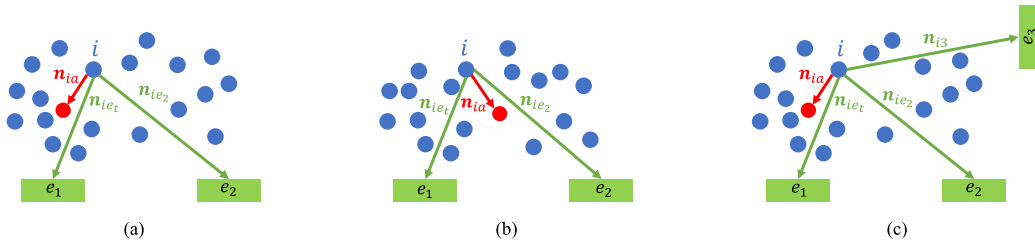


Fig. 5. A diagram of the exit-reorientation mechanism.

are attractive. Lakoba [29] also believes that exits have a particular attractiveness, which leads people to move toward the space outside the room. The formula of attraction force is shown in Eq. (12), where s represents the joint exit and $\mathbf{n}_{is}$ is the unit vector of pedestrian i pointing to joint exit s . A_s and B_s are constants. r_i and d_{is} represent the body radius and the distance from pedestrian i to the exit s , respectively. Escaping to low risk areas provides some incentive function for pedestrians because it represents hope to a certain extent. Under the effect of incentive, the proverb “Escape from the Tiger’s Mouth” often occurs. Adding the model’s social force of “low risk area attraction” can effectively reproduce this phenomenon.

$$\mathbf{F}_{is} = A_s \exp \left[(r_i - d_{is}) / B_s \right] \mathbf{n}_{is} \quad (12)$$

3. Simulation results and analysis

The proposed model is tested in the space of 39 m × 29 m with two exits that imitates a ticket hall. Pedestrians and attackers are initially distributed randomly before a test. The random distribution does not have a significant impact on the results. The meanings of circles with different colors are as follows: blue circles represent pedestrians, red circles represent attackers, and the color of the pedestrians changes based on the extent of injury after an attack. Escaping from the hall through one of the exits is considered a successful escape. The movement and casualty rules of pedestrians are shown in Fig. 6. Here, d_{attack} is the critical attack distance pedestrians might be injured. This paper selects attack tools, critical attack distance d_{attack} , and casualties probability of simulations by a literature review as shown in Table 1. Given that the attacker used a hacking knife in “3.1 Kunming Railway Station Violent Terrorism Case”, the probabilities of minor injury, serious injury, and death in simulations are set to 0.0625, 0.1875, and 0.75 respectively, according to Table 1 [30]. Minor injuries, severe injuries, and dead pedestrians are represented by light gray, gray, and black circles in simulations, respectively. The attack rule of attackers is shown in Fig. 7. Here, t_{attack} is an attack interval. In terms of parameter selection, the range of values for $d_{observe}$ is determined by the scenario’s size. The setting of d_{act} was based on

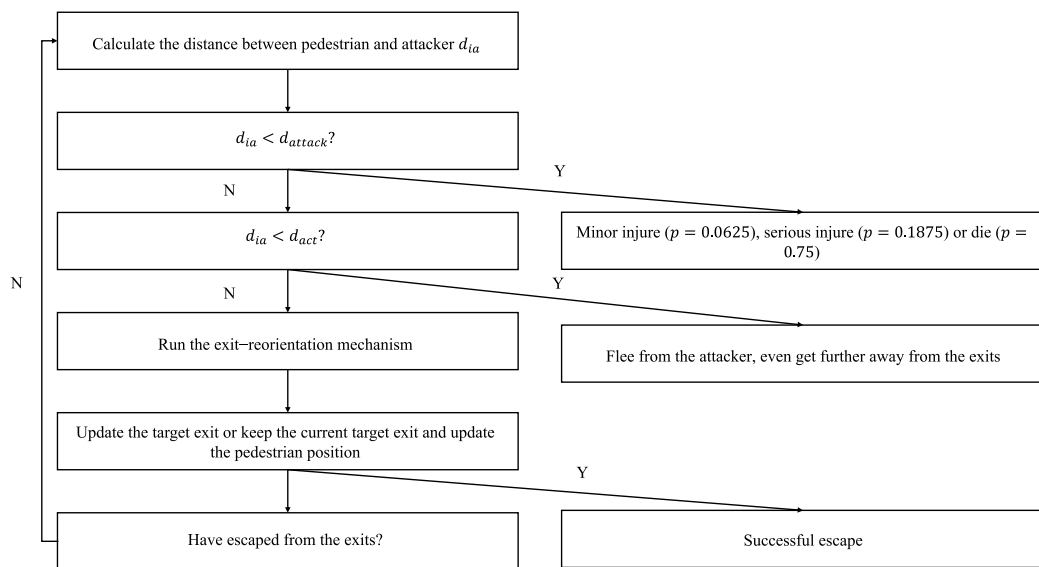


Fig. 6. The movement rule and casualty rule of pedestrians.

Table 1
Parameters of three kinds of sudden attacks.

Attack tools	Attack distance (m)	Probability of minor injury	Probability of serious injury	Probability of death
Dagger	0.5	0.5625	0.1875	0.25
Kitchen knife	0.7	0.2025	0.2475	0.55
Hacking knife	1	0.0625	0.1875	0.75

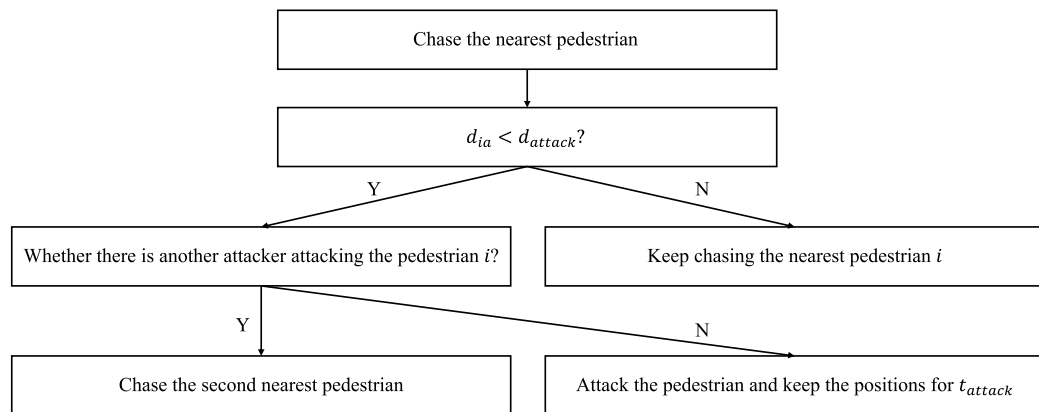


Fig. 7. The attack rule of attackers.

Lu's research [31]. The values of parameters A_s and B_s take into account the need for the magnitude of attractiveness to be of the same order of magnitude as other forces. The parameter settings during simulation are shown in Table 2. In the following sections, we conduct several simulative experiments to explore the effects of observation distance $d_{observe}$, risk tolerance level d_{act} , and the ratio of rational pedestrians r etc. on evacuation.

3.1. Model validation

The improved model is validated quantitatively and qualitatively in this part, respectively. First, we validate the model qualitatively by comparing particular evacuation phenomena between the snapshots of the '3.1 Kunming Railway Station Violent Terrorism Case' video and snapshots of simulations in three dimensions as shown in Fig. 8. First, the attack causes local congestion, as Fig. 8(a) and (b) show. The sudden attack increases the local density of the nearby crowd, and the improved model reproduces this

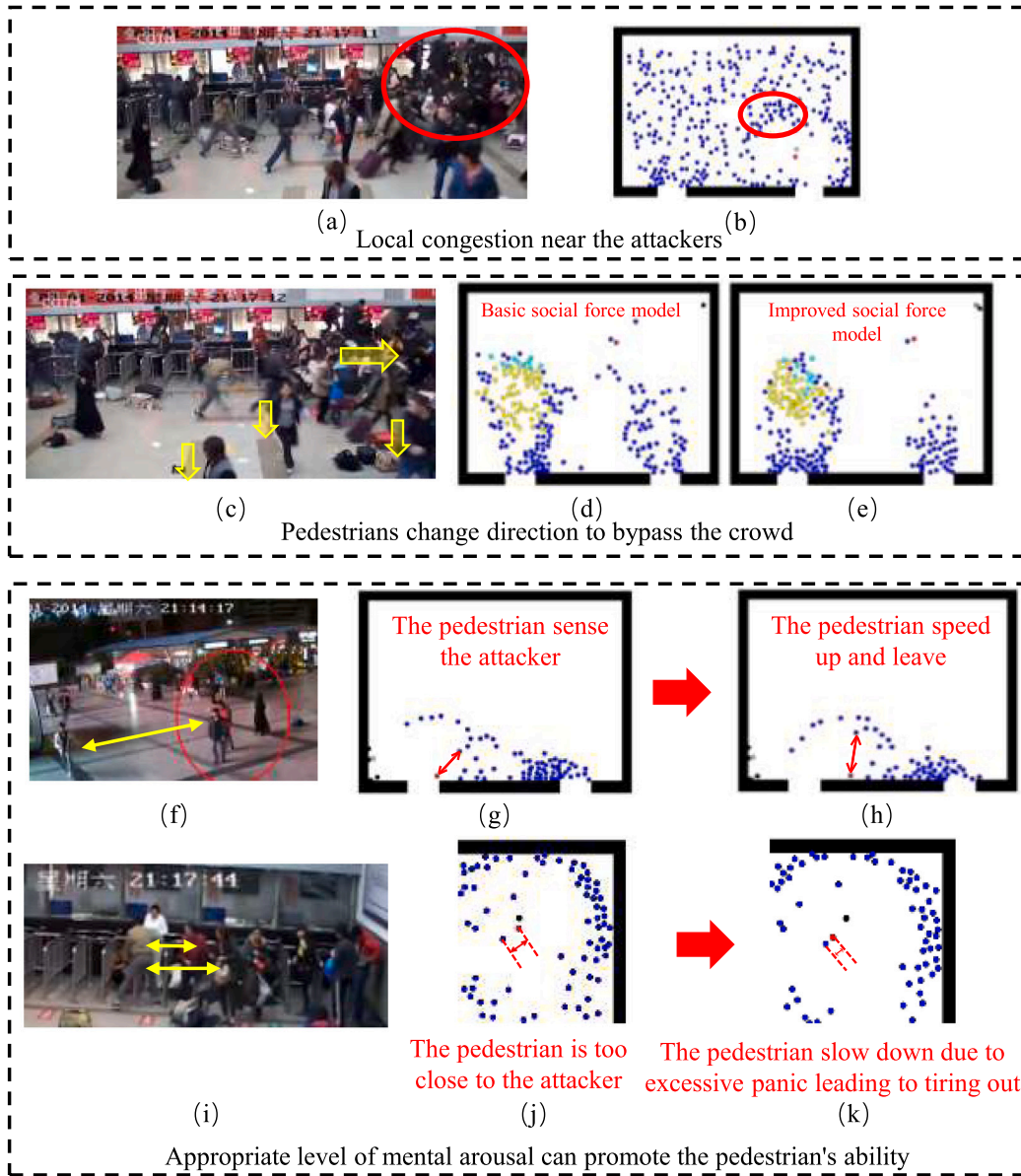


Fig. 8. Phenomena comparing between video snapshots and simulation snapshots.

phenomenon well. Second, if there is congestion ahead, not everyone will choose to queue but turn and run in the other direction to bypass the crowd. Fig. 8(e) shows the phenomenon that people (blue-green circles) bypass the crowd by changing the desired direction. Third, a phenomenon consistent with the Yerkes-Dodson law is observed: pedestrians accelerate within a certain distance from the attacker but slow down or do not escape when the distance is too close. The snapshot like Fig. 8(f) shows that the man begins to flee from attackers when the distance between them is not too far or close. However, in Fig. 8(i), some people stop and watch instead of running. The matching parts are marked with circles in the figure, respectively.

Then we validate the model quantitatively by comparing fundamental diagrams between common conditions [32], video data of gunshot [33], and our data in Fig. 9. The fundamental diagrams of common conditions refer to the relationship between pedestrian density and velocity calculated through video analysis, observation on location, or observation of photographs of unidirectional or bidirectional pedestrian flow under non-emergency conditions by researchers such as Helbing and Weidmann et al. In Fig. 9, the light blue diamond-shaped data points represent the average instant speed of all pedestrians and the average density in a specific area during evacuations of a shooting attack at an Apple Store located in a New York mall in 2016. We count the average density and speed in a 20 m^2 area near one of the exits. It shows a big difference between common conditions and sudden attacks in the

Table 2
Parameters' interpretations and values.

Parameters	Interpretations	Value
m_i	The mass of pedestrian i	80 kg
τ_i	Characteristic time	0.5 s
a	The contribution of the realization degree of desired speed to panic	1/3
b	The contribution of the proximity to the attacker to panic	1/3
c	The contribution of the crowdedness to panic	1/3
$v_i^{\min}$	Minimum speed of pedestrian	1.5 m/s
$v_i^{\max}$	Maximum speed of pedestrian	5 m/s
$\rho_{\text{immovable}}$	The critical density that pedestrians are too crowded to move	2.3 /m ²
d_{attack}	The critical attack distance pedestrians might be injured	1 m
t_{attack}	The attack interval	1 s
A_s	Strength of psychological repulsion	240
r_i	Radius of the pedestrian	0.3 m
B_s	Scope of psychological repulsion	15

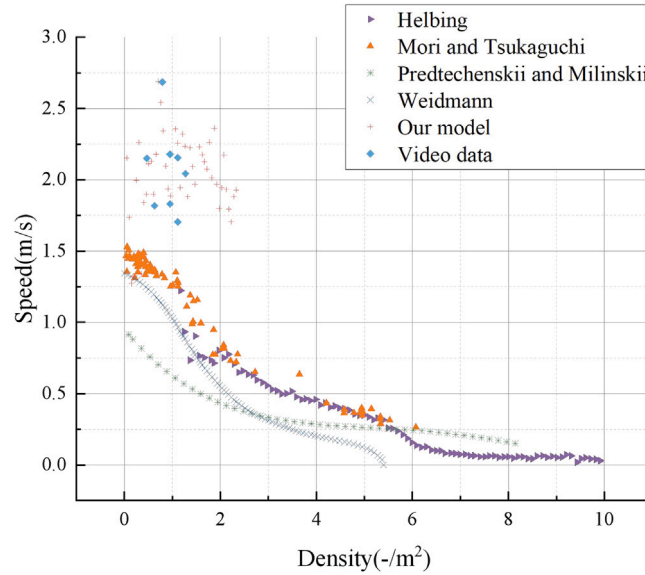


Fig. 9. Fundamental diagrams comparing between previous studies [32,33] and our data.

relationship between speed and density. Our data fit well with existing gunshot video data and has more data points. It proves that our model is relatively realistic in modeling pedestrian behaviors.

In addition, we compare the simulation results with related studies. We first compare our data to those of Lu's research [31]. Lu's and our results have a similar trend regarding the effect of the number of attackers on the survival rate. The survival rates range from roughly 60% to 95% when the number of attackers ranges from more to less. Furthermore, the survival rates with 1 or 3 attackers match Lu's results. Fig. 10 is the comparison of the results, where (d) is the result of Lu's study. (a), (b), and (c) represent the survival rate dynamics at population sizes of 100, 200, and 300 individuals, respectively. (d) represents the survival rate dynamics of 1000 individuals simulated by Lu using NetLogo software. It proves the credibility of the results to a certain extent.

We also compare casualties between Liu's data [18] and ours as shown in Fig. 11. We compare the number of casualties at $n = 1, 2, 3, 4$ and $N = 100, 200, 300$ respectively. It shows that the casualties when there is one attacker are roughly the same. However, with the increasing number of attackers, the difference between Liu's data and ours is becoming increasingly obvious. Regarding this increasingly apparent distinction, it is highly likely to be a result of pedestrians' choices of low-risk exits. Excessive selection of the same exit by pedestrians can cause congestion at the exit in a short period. If an attacker moves toward this exit at that time, significant casualties may occur due to insufficient time for evasion. Additionally, another potential reason could be the negative impact of psychological states, resulting in more casualties among pedestrians during the escape process compared to the casualties in Liu's simulation.

Then the three parameters' sensitivities are analyzed, which are a , b , and c , respectively. They are positive constants between 0 and 1, and $a + b + c = 1$. We fixed one parameter at 1/3 and allowed the other two parameters to vary within the range [0, 2/3] to explore how their variations affect the evacuation time. Fig. 12 shows the simulation results. Each point in the figure was simulated 50 times. Fig. 12 (a) shows the variation in evacuation time as a increases, b decreases and c remains constant. Fig. 12 (b) shows the variation in evacuation time as a increases, c decreases and b remains constant. It can be observed that when c and b are constant,

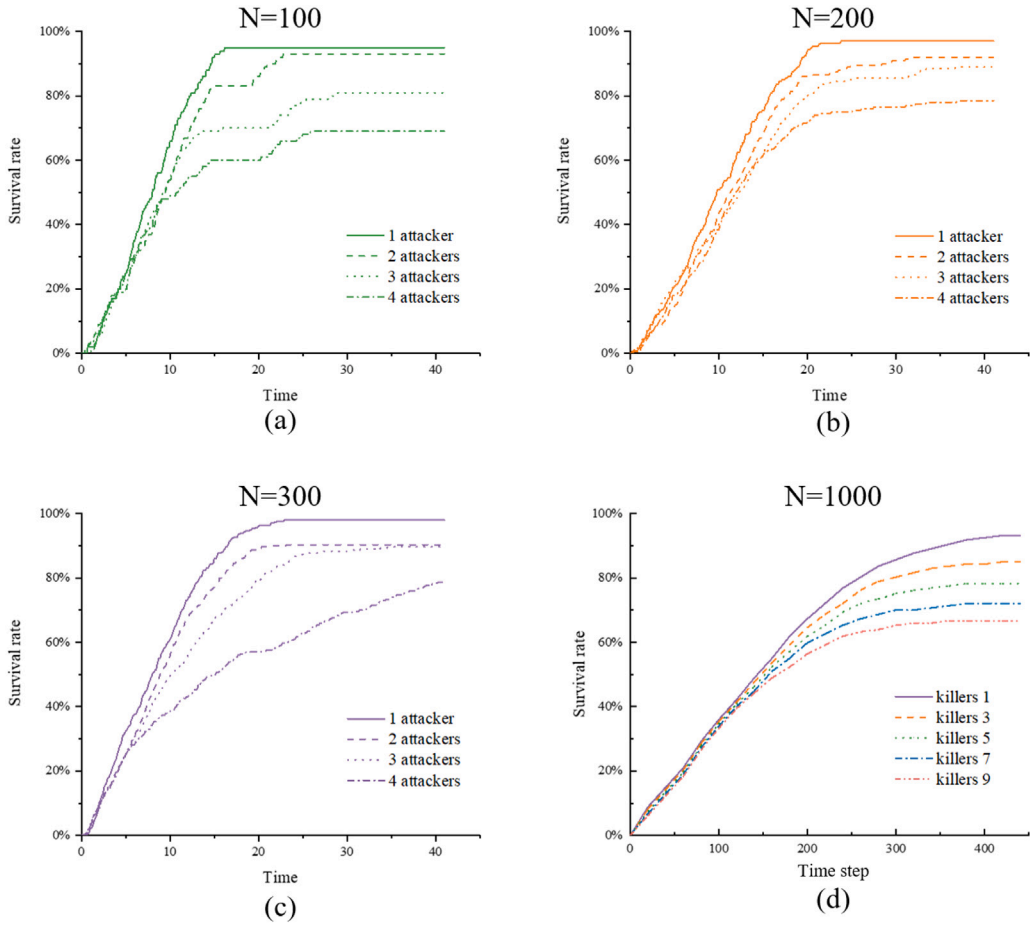


Fig. 10. The comparison of survival rate between related studies and ours.

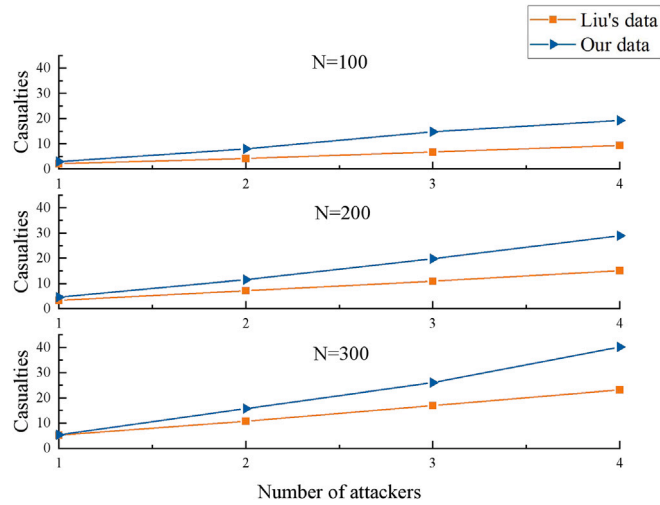


Fig. 11. The comparison of casualties between related studies and ours.

parameter a reaches its minimum evacuation time at $2/9$ and $1/3$, respectively. Meanwhile, when a is kept constant, b and c have minimal impact on the results.

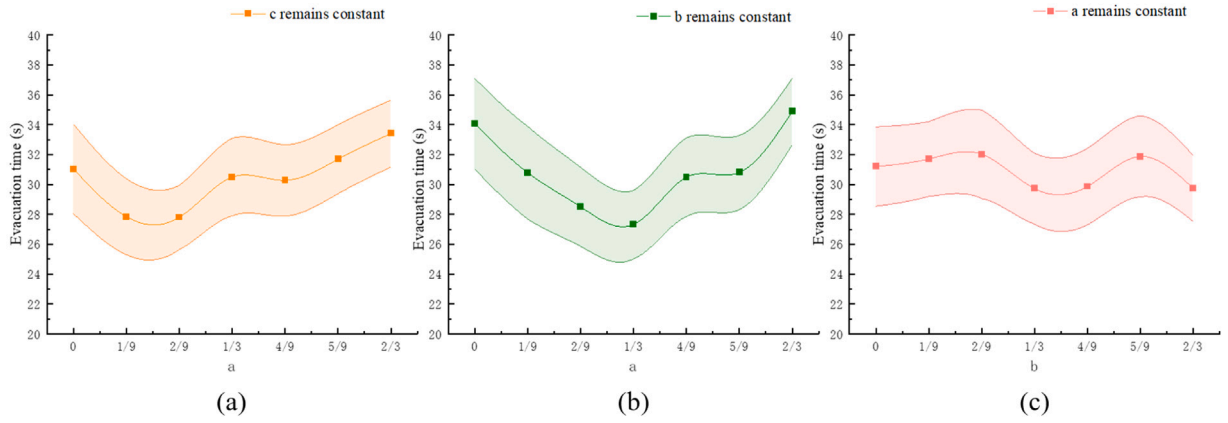
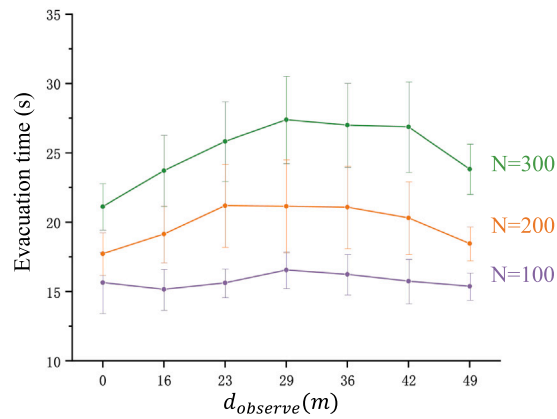
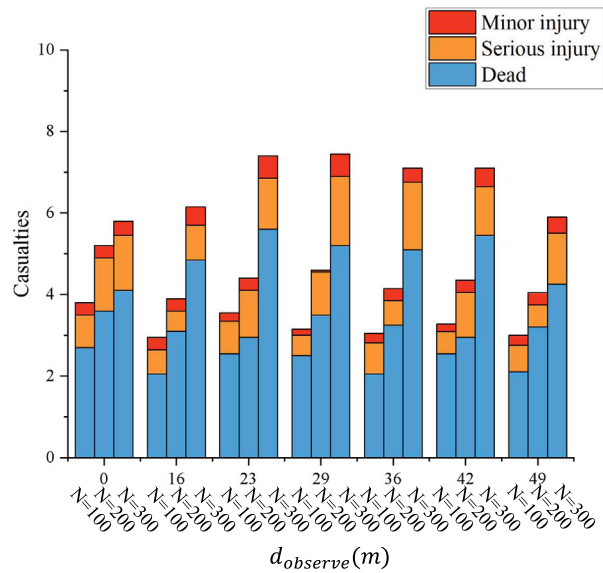


Fig. 12. Parameter sensitivities.

Fig. 13. The effect of $d_{observe}$ on evacuation time.Fig. 14. The effect of $d_{observe}$ on casualties.

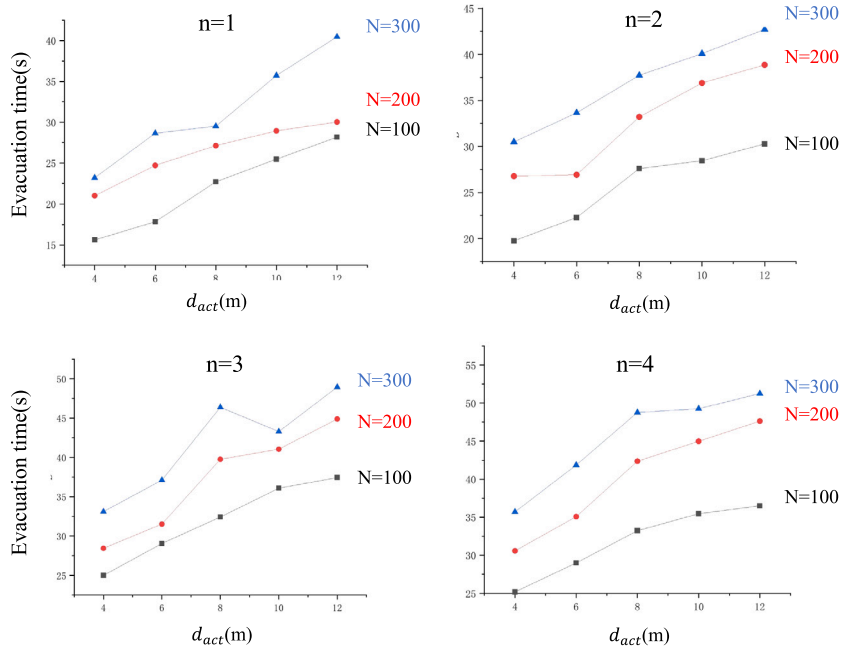


Fig. 15. The effect of d_{act} on evacuation time.

3.2. Effect of environmental visibility, risk tolerance level and rational pedestrians

3.2.1. Effect of the observation distance $d_{observe}$

This part of the research holds greater significance for attacks accompanied by smoke grenades. As an important factor in the exit-reorientation mechanism, $d_{observe}$ is the observation distance within which pedestrians can judge the relative positions of attackers and themselves. Environmental visibility also determines it. It will likely impact the time to reorientate the target exit and the pedestrians' distribution at two exits. Because of a certain degree of randomness of the simulations, we choose to perform multiple simulations and take the average. We conduct 420 simulations altogether in this section in which $d_{observe}$ is taken as 0 m, 16 m, 23 m, 29 m, 36 m, 42 m and 49 m while N is taken as 100, 200 and 300 respectively.

We investigate the effect of $d_{observe}$ on evacuation time with 1 attacker for different numbers of pedestrians, as shown in the interval graph Fig. 13. The results show an obvious difference between the three kinds of simulations in evacuation time. However, they have almost the same trend with the change of $d_{observe}$. The evacuation time increases when $d_{observe} < 29$ m, and decreases when $d_{observe} > 29$ m with 100 pedestrians and 300 pedestrians. The maximum evacuation time is achieved at $d_{observe} = 23$ m for 200 pedestrians. The possible reason is that the premature response of pedestrians based on the exit-reorientation mechanism makes more people gather at one exit, thus making the evacuation longer. According to Fig. 14, $d_{observe}$ also influences casualties in evacuation. It shows the number of minor injuries, serious injuries, and deaths, respectively, in three kinds of pedestrian number settings with different $d_{observe}$. The maximum casualties are achieved when $d_{observe} = 23$ m, 29 m for 300 pedestrians. Nevertheless, no obvious trend in casualties changes with $d_{observe}$ for 100 or 200 pedestrians. Therefore, longer evacuation time can increase the number of casualties. The trend is evident when the number of pedestrians N exceeds a threshold between 200 and 300.

3.2.2. Effect of the minimum distance for risk tolerance d_{act}

d_{act} is a parameter that describes pedestrians' risk tolerance. Fig. 6 illustrates how it affects the evacuation process. To investigate the effect of d_{act} on evacuation time, survival rates, and casualties, we conduct several kinds of simulations with 1, 2, 3, and 4 attackers and 100, 200, and 300 pedestrians from $d_{act} = 4$ m to $d_{act} = 12$ m. We conduct 1320 simulations in this section in total. The results are shown in Figs. 15–17. Here, n is attackers' number, and N is pedestrians' number. The folded lines of Fig. 15 almost have critical points at $d_{act} = 8$ m. When $d_{act} > 8$ m, the growth rate of evacuation time slows down. Therefore, excessive distance for risk tolerance could make an even less smooth evacuation process. However, the next section will investigate whether the proportion of rational people will affect evacuation time. The effect of d_{act} on survival rates is shown in Fig. 16. The survival rates of all kinds of simulations with different numbers of pedestrians have a smooth stage between $d_{act} = 4$ m and $d_{act} = 6$ m. It proves that when $d_{act} \leq 6$ m has a negligible effect on survival rates. The folded lines of 200 and 300 pedestrians' simulations in Fig. 16 have almost the same trend. Interestingly, even though the casualties increase as the crowd becomes denser, so does too the survival rate overall which is also observed in O. Abreu's research [34]. The folded lines of survival rates with 100 pedestrians have an evident decreasing trend when $d_{act} > 6$ m no matter how many attackers are. Therefore, the change in survival rates with

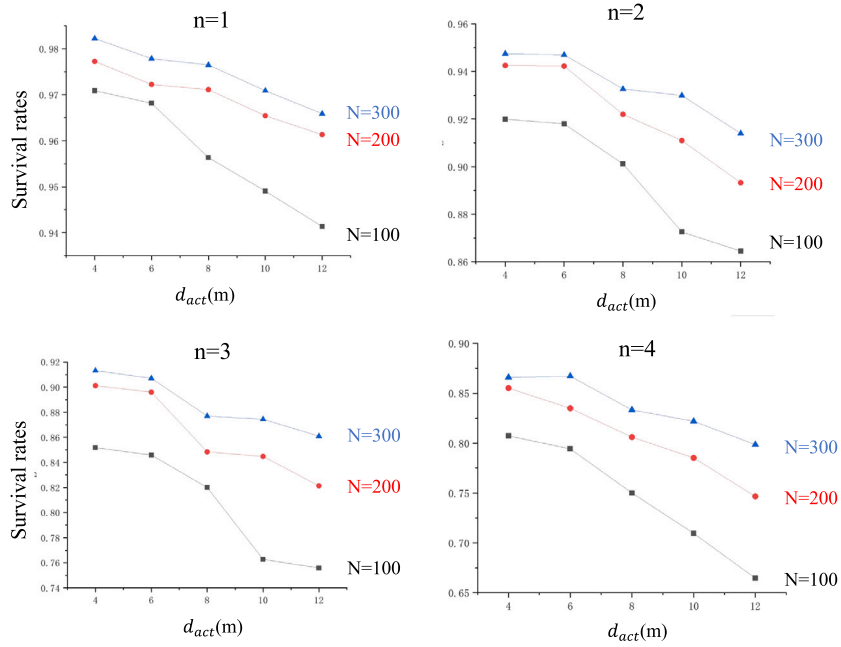


Fig. 16. The effect of d_{act} on survival rates.

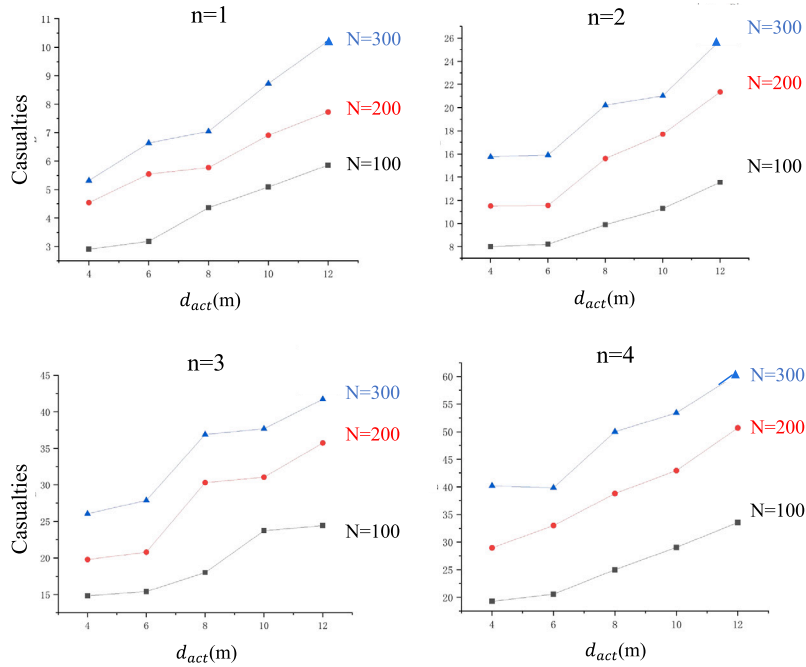


Fig. 17. The effect of d_{act} on casualties.

pedestrians' numbers is not linear. Fig. 17 shows the effect of d_{act} on casualties. Interestingly, the folded lines' trend in Fig. 17 is similar to the trend of folded lines in Fig. 15 respectively, especially when $n = 1$ and $n = 3$. The simultaneous increase in casualties and evacuation time due to the attack is a possible reason for the similar trend. According to this, the impacts of the attack process on evacuation time and casualties are significant.

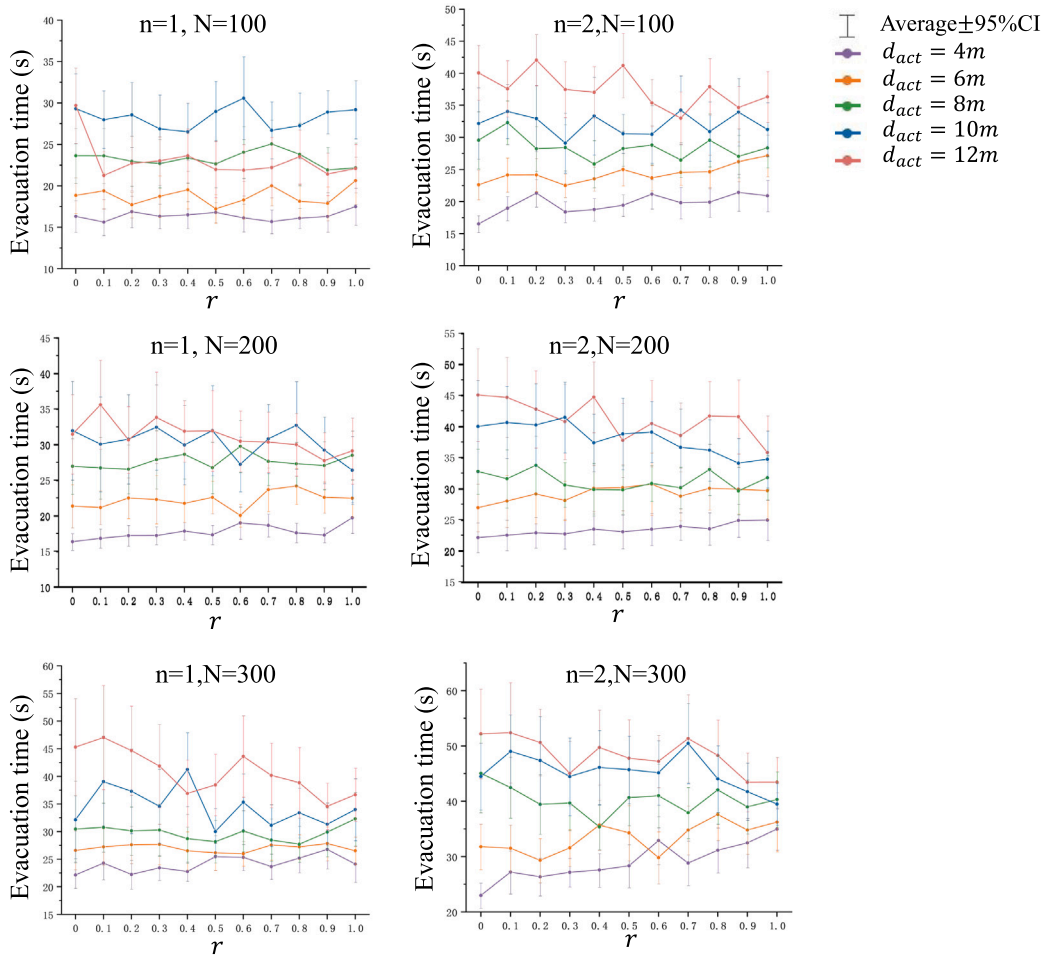


Fig. 18. The effect of the ratio of rational pedestrians r on evacuation time when $n = 1, 2$.

3.2.3. Effect of the ratio of rational pedestrians r

The proposed model considers the positive and negative effects of environmental information on pedestrian's psychology. It also considers the weakening of panic-based repulsive forces. However, not every pedestrian could escape rationally. Some irrational people might escape toward the nearest exit throughout the whole process. Therefore, the different ratios of irrational pedestrians or rational pedestrians could impact the congestion of one of the exits and then affect the evacuation time. This part will investigate the effect of the ratio of rational pedestrians on the escape process. The number of attackers $n = 1, 2, 3, 4$. The number of pedestrians $N = 100, 200, 300$. The minimum distance for risk tolerance of rational pedestrians $d_{act} = 4\text{ m}, 6\text{ m}, 8\text{ m}, 10\text{ m}, 12\text{ m}$. The ratio of rational pedestrians $r = 0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0$. We conduct 660 groups of simulation experiments. Each group is executed 22 times. 14,520 simulations are performed in this section in total. Figs. 18 and 19 are interval graphs showing the evacuation time changes with the ratio of rational pedestrians r .

A shrinkage trend of evacuation time is observed by comparing the results of 4 kinds of the number of attackers. It means the value range of evacuation time reduces with the ratio r . This trend is not obvious when $n = 1$ and $N = 100$. It is probably due to the few pedestrians participating in the escape, and the outlet flows smoothly. Except for this case, the slope trend of the other evacuation time curves changes from positive to negative with d_{act} from 4 m to 12 m. Moreover, the slope trend of curves almost reaches 0 when $d_{act} = 8\text{ m}$. That is to say, the evacuation time is in an uptrend with the increasing ratio r when $d_{act} < 8\text{ m}$. Similarly, the evacuation time is in a downtrend with the increasing ratio r when $d_{act} > 8\text{ m}$. Therefore, the larger the ratio of rational pedestrians r , the shorter the evacuation time is likely to be when $d_{act} > 8\text{ m}$. The larger the ratio of rational pedestrians r , the longer the evacuation time is likely to be instead when $d_{act} < 8\text{ m}$. Fig. 20 shows the evacuation snapshots when $r = 0, 0.5, 1.0$ and $d_{act} = 4\text{ m}, 8\text{ m}, 12\text{ m}$. The evacuation time of each simulation is on the top of subfigures. The results corresponding to $r = 0$ are generated by the simulations with basic model without improvements. An appropriate value of r could improve pedestrians' choice of low risk exits. However, it might cause a slower or a faster escape which is determined by the attacker's position. The evacuation time has a more obvious increasing trend with the number of pedestrians N than that with the number of attackers n .

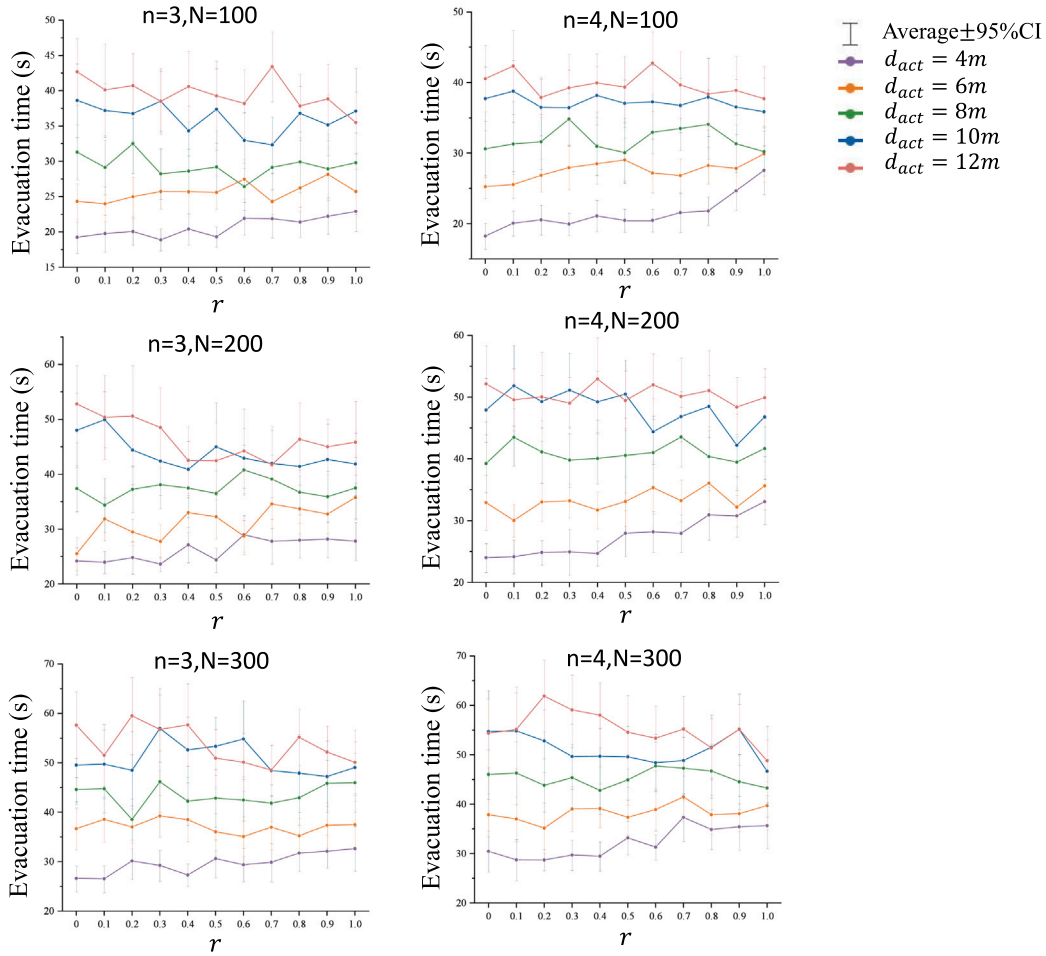


Fig. 19. The effect of the ratio of rational pedestrians r on evacuation time when $n = 3, 4$.

4. Conclusions

In this paper, we propose and validate the improved model quantitatively and qualitatively. Then we conduct three simulations to investigate the effect of environmental visibility, pedestrian risk tolerance level, and the ratio of rational pedestrians on evacuation with the improved model in a simple scenario. Reviewing the entire study process and integrating the analysis of various data, the following conclusions, which correspond to the simple scenario, are summarized: (1) The evacuation time is longer when $d_{observe} = 23$ m or $d_{observe} = 29$ m. The reason might be that pedestrians choose the same exit, which has low risk together at a certain moment, resulting in congestion at that exit and thus increasing the evacuation time. Therefore, it is important to enhance the evacuation guidance to allocate the exits for pedestrians and improve $d_{observe}$ by broadcasting the risk location under emergencies. (2) The maximum casualties are achieved when $d_{observe} = 23$ m, 29 m for 300 pedestrians. Furthermore, the effect of $d_{observe}$ on casualties will become more obvious after the number of people exceeds a threshold between 200 and 300. (3) The larger the d_{act} , the slower the evacuation tends to be. However, the effect of increasing d_{act} on evacuation time becomes significantly weaker when $d_{act} > 8$ m. (4) d_{act} has a small effect on survival rates when $d_{act} \leq 6$ m. The survival rates with 100 pedestrians have a more evident decreasing trend when $d_{act} > 6$ m compared with the scenarios with 200 or 300 pedestrians. (5) The larger the ratio of rational pedestrians r , the shorter the evacuation time is likely to be when $d_{act} > 8$ m. The larger the ratio of rational pedestrians r , the longer the evacuation time is likely to be instead when $d_{act} < 8$ m.

Our work innovatively integrates the social force model with Yerkes-Dodson law to make it applicable to simulate pedestrian evacuation under sudden attack. It also makes the behavior of pedestrians in a state of panic more convincing. At the same time, it will help people better understand how the heterogeneity of psychological states can affect evacuation efficiency. This study has some limitations. Since only videos and others' data are used for validation without actual evacuation data, this verification will be one of the future works to be completed. Furthermore, the ability of the improved model to be applied in more complex scenarios will be further demonstrated in future work.

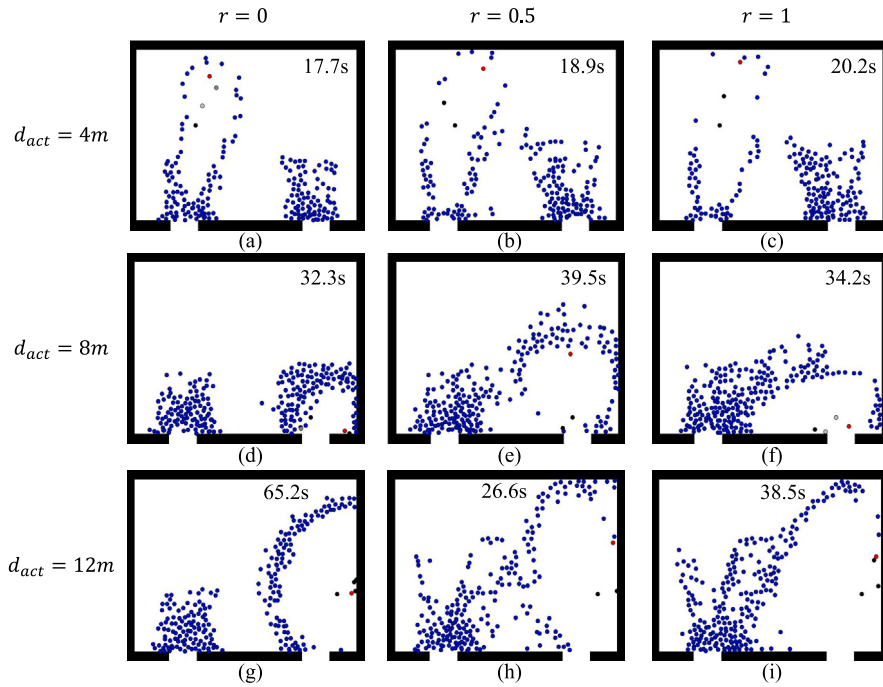


Fig. 20. The simulation snapshots with the settings of $n = 1$ and $N = 300$.

CRediT authorship contribution statement

Jiaming Liu: Methodology, Software, Validation, Writing – original draft. **Hui Zhang:** Writing – review & editing, Supervision, Project administration. **Ning Ding:** Conceptualization, Writing – review & editing, Funding acquisition. **Yuntao Li:** Resources, Visualization, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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